

Code-Layout, Readability and Reusability

Date	28 June 2026
Team ID	
Project Name	EduGenie
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the EduGenie codebase by assessing its structure, readability, maintainability, and reusability. A modular design, consistent coding practices, and well-organized components ensure that the application remains scalable, easy to maintain, and adaptable for future enhancements.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout	Yes	Standard 4-space PEP8 indentation followed across Python files.
2	Proper File Structure	Files and folders are logically organized	Yes	Project modularized into app/, routes/, services/, static/, and templates/.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Clean snake_case variable names reflecting their purpose.
4	Function / Method Names	Functions are descriptively named	Yes	Descriptive function names explaining the business logic.
5	Code Comments	Inline and block comments explain logic	Yes	Docstrings and inline comments detailing AI prompt construction.
6	Modular Design	Code is split into reusable functions/modules	Yes	Logical separations in services/ (auth, AI models, and RAG).
7	No Redundant Code	Duplicate or unused code is removed	Yes	No duplicate code; shared helper utilities used.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Global FastAPI exception handler handles server errors gracefully.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
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1	ai_service (Gemini wrapper)	Python	Shared by Q&A, explain, quiz, roadmaps	High
2	pdf_exporter	Python	Used by quiz and summary export modules	High
3	auth_middleware	Python	Used across all private API routes	Medium
4	flat_file_db_base	Python	Inherited by users, quizzes, summaries JSON files	High
5	text_extractor	Python	Used by PDF and DOCX file processing in RAG	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Excellent structure following FastAPI design patterns.
Readability	5	Highly readable code with consistent naming conventions.
Reusability	5	Highly modular helper files in app/utils/ and app/services/.
Documentation / Comments	5	Fully documented routes and helper classes.
Overall Score	5	Clean, production-ready codebase.